

Rhianna Ruggiero

DSC 640

Week 7 & 8 Assignment – Lottery Numbers

Audience:

The audience for this story will be my coworkers. They will not be familiar with the data surrounding the lottery and they will be coworkers at analyst or senior analyst positions.

Purpose:

The purpose of this story is to convince my coworkers to pool money together to purchase many lottery tickets to increase our odds of winning. I want to show them information about the past winning numbers and to show previous numbers that were picked more often and other previous trends that could potentially be good choices for the numbers that we pick. I am essentially asking for approval for my coworkers to join me in a group purchase.

Medium:

I will present the information using a PowerPoint since I will set up a meeting with my coworkers to show them the information during a work day lunch. I believe this will be the best way to show the information since my coworkers are used to seeing presentations with visuals on them and it is a good way to show a few people in one room this information.

Design:

For each lottery I kept the colors similar so that the audience would know which charts go with which lottery more easily. I also grouped the lotteries together so that I would not flip between different lotteries in the presentation and confuse the audience. I also avoided using reds and greens on one chart together in case anyone was color blind. I added in a legend when necessary to provide more information for some charts like the bubble chart. I also made sure that each visual had a title that was aligned at the top left of the chart and also made sure that there were axis labels on graphs as well.

Ethical Considerations:

There are some regulatory guidelines regarding the lottery in that everyone must be 18 years or older to purchase a lotto ticket. Only one person may claim the prize which could lead to issues with taxes and would need to be figured out with an expert so that the money gets distributed evenly between winning parties. The data is credible and was sourced ethically in that it was collected from the official New York data government site. For each data set I had to pivot the winning numbers data so that I could determine averages and treat each number chosen equally. The lottery is random and seeing which numbers were picked more often in the past does not

guarantee that they will continue to be chosen often in the future so it should be obvious during my presentation that choosing specific numbers will not necessarily increase the odds of winning. While analyzing the data I assumed that every winning number for each specific lottery was equally important and that specific ranges of numbers didn't have more weight than others.



Pitch in to buy Lotto Tickets with me!

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The lotteries:

Pick 10 Lotto Numbers

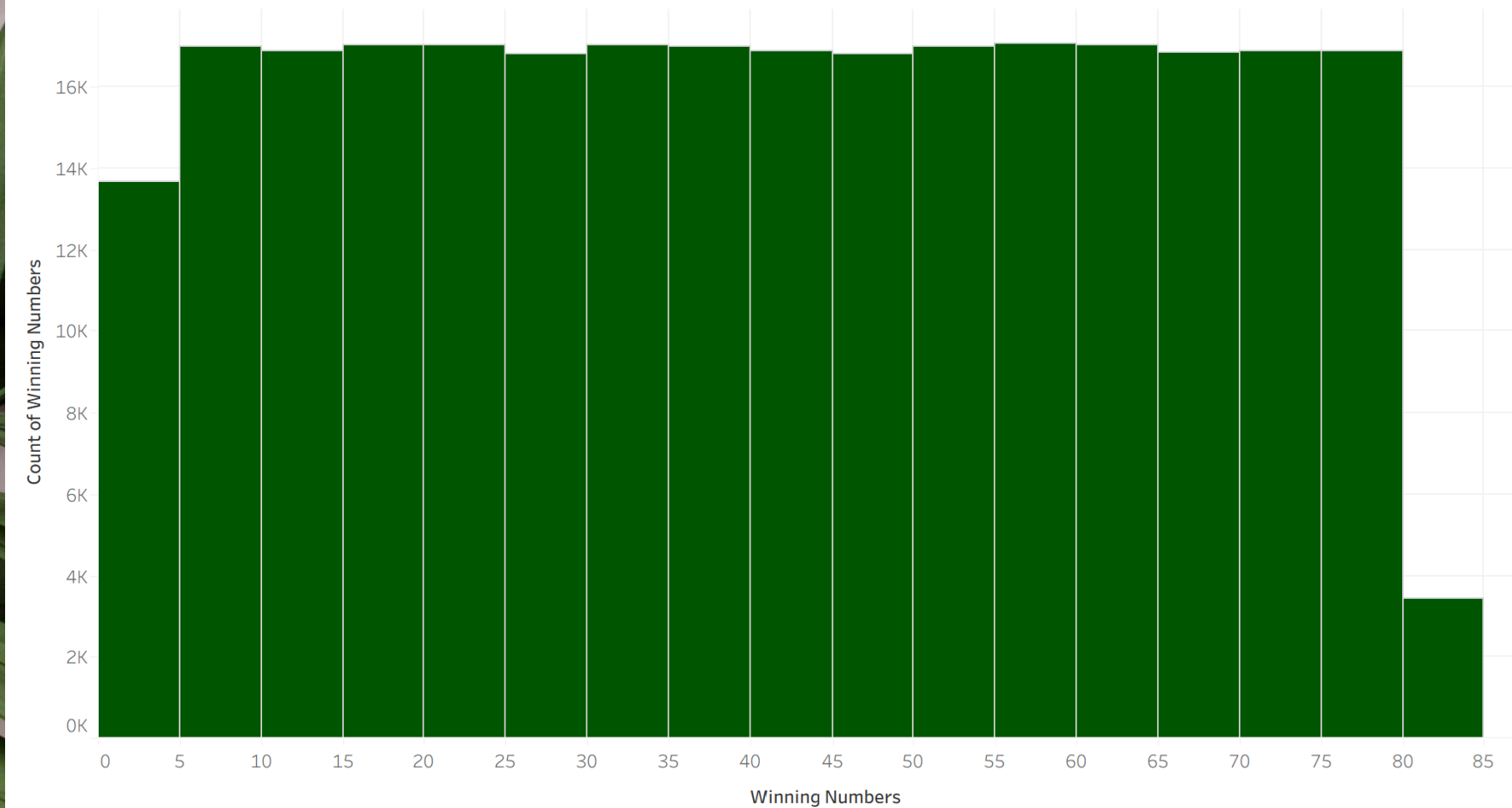
MegaMillions Lotto Numbers

Powerball Lotto Numbers

Pick 10 Lotto Numbers

- The pick 10 lottery has an even distribution of winning numbers, but less numbers seem to be under 5 or above 80
- Any tickets we buy together can focus less on numbers below 5 or above 80

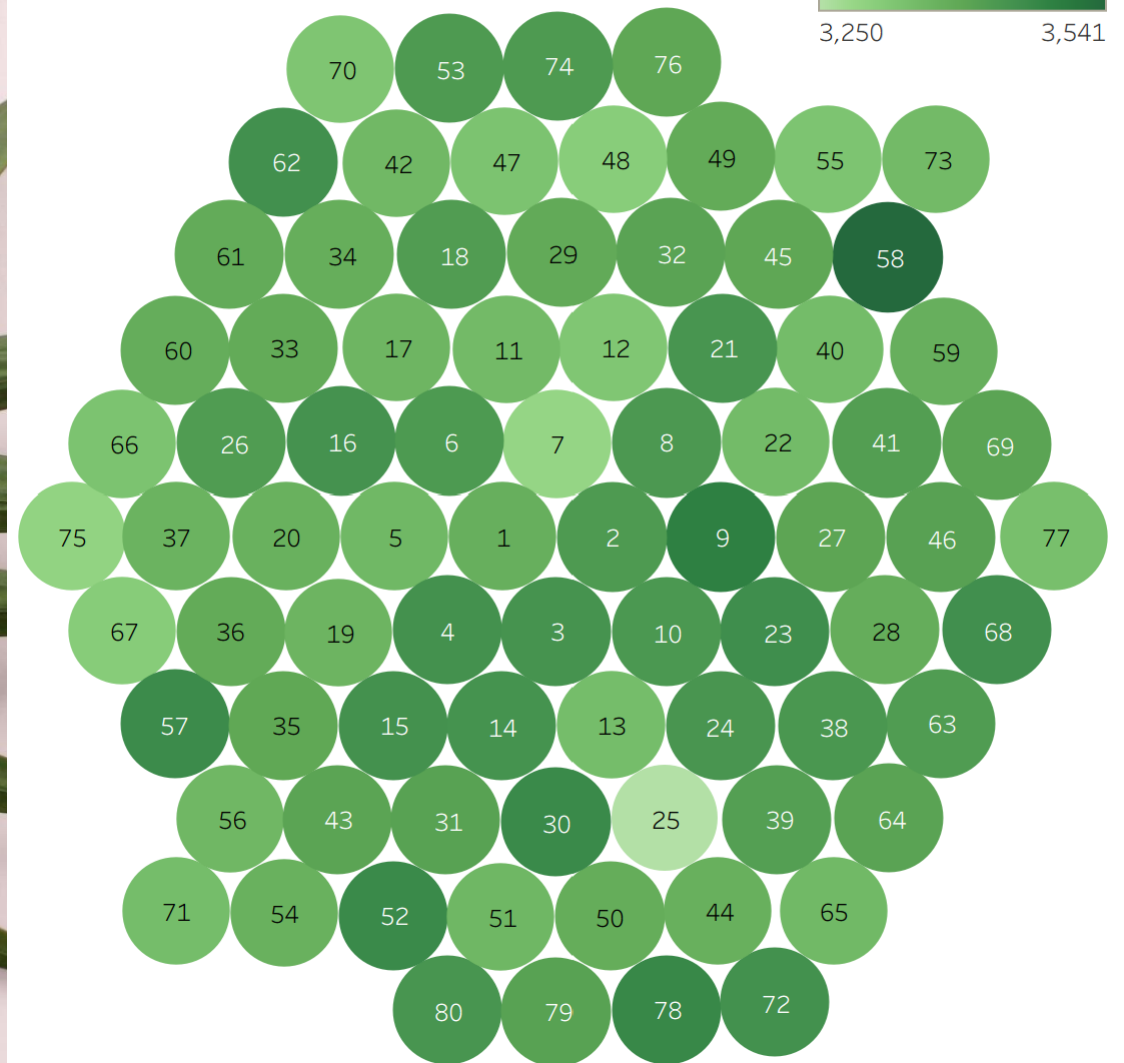
Frequency of Winning Numbers



Pick 10 Lotto Numbers

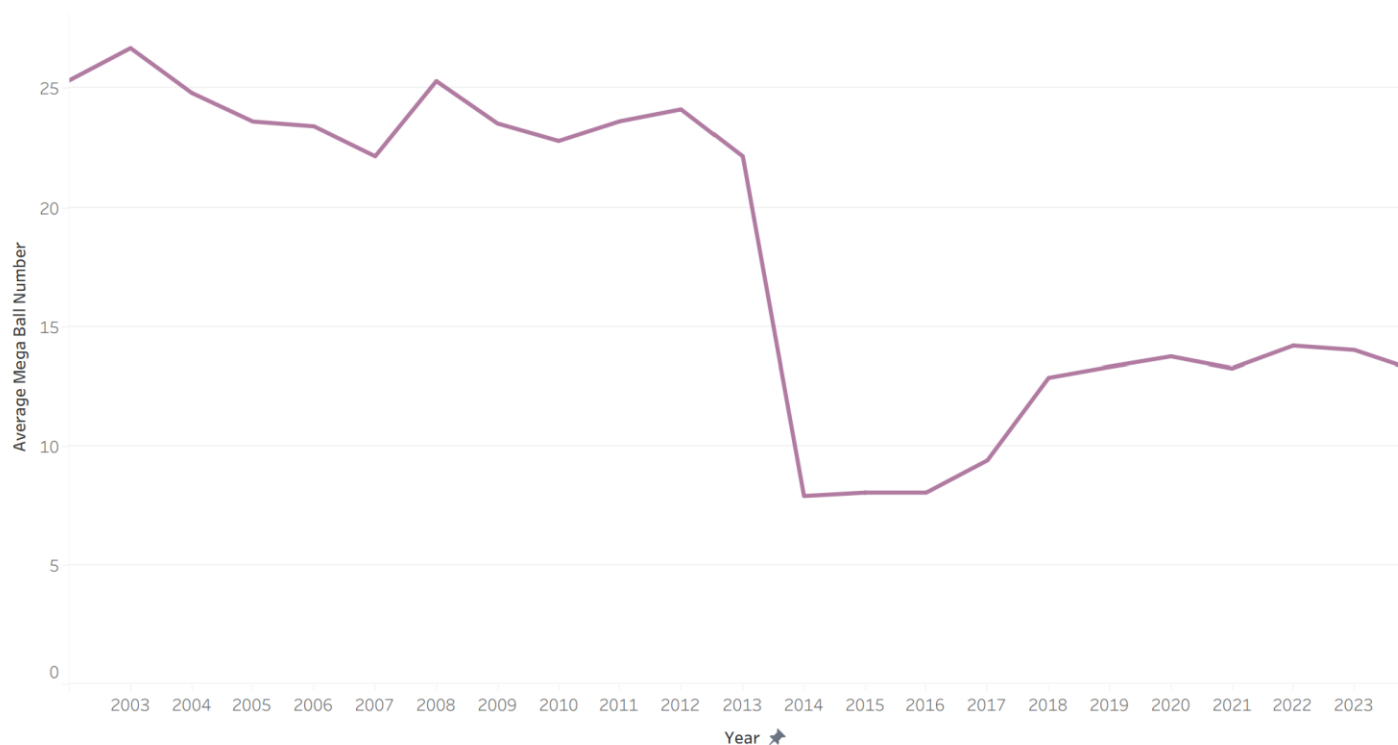
- The most frequently chosen winning number is 58
- The less frequently chosen number is 25
- By pooling our money together, we can buy more pick 10 lotto tickets that have number 58 in them to hopefully increase our odds of winning together

Most Frequently Chosen Numbers



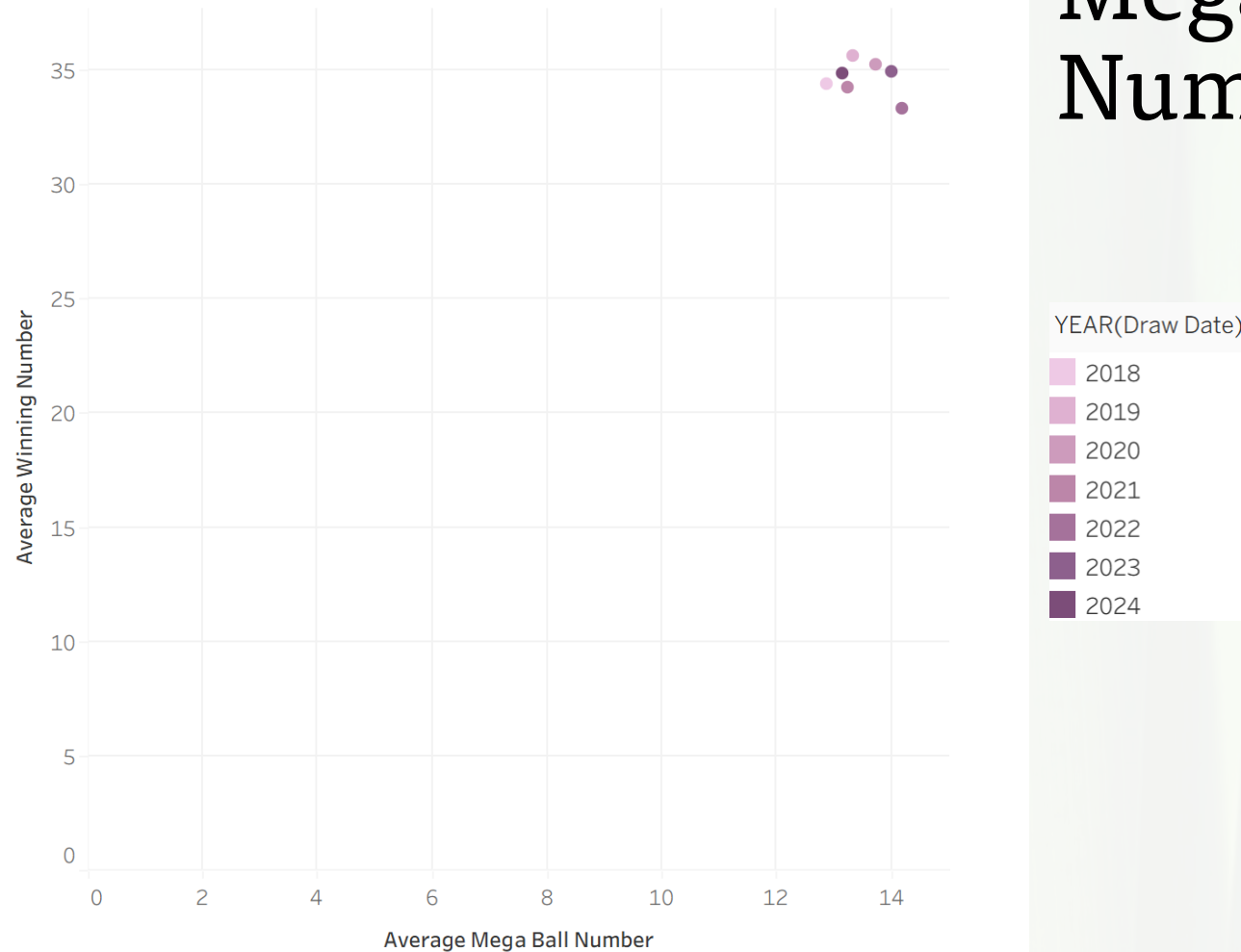
MegaMillions Lotto Numbers

Average Mega Ball Number Over Time



- On average the mega ball number has decreased over time and recently has on average been between 10 and 15.
- If you join the pool with me, we can choose tickets with mega ball numbers between 10 and 15 near the average

Average Winning Numbers and Mega Ball Number since 2018



MegaMillions Lotto Numbers

- The average mega ball number in the past few years has been around 12-14
- If we purchase tickets together, we can try and choose a mega ball number in this range
- By pooling our money together, we can buy more tickets for less money per person with mega ball numbers 12, 13 and 14
- There doesn't seem to be a relationship between the average winning number and the average mega ball number, however in each year the average winning number seems to be around 34 or 35 which we could use to help us choose our numbers

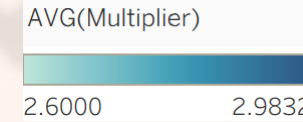
Powerball Lotto Numbers

Count of Winning Numbers Below 36 from 2021-2023



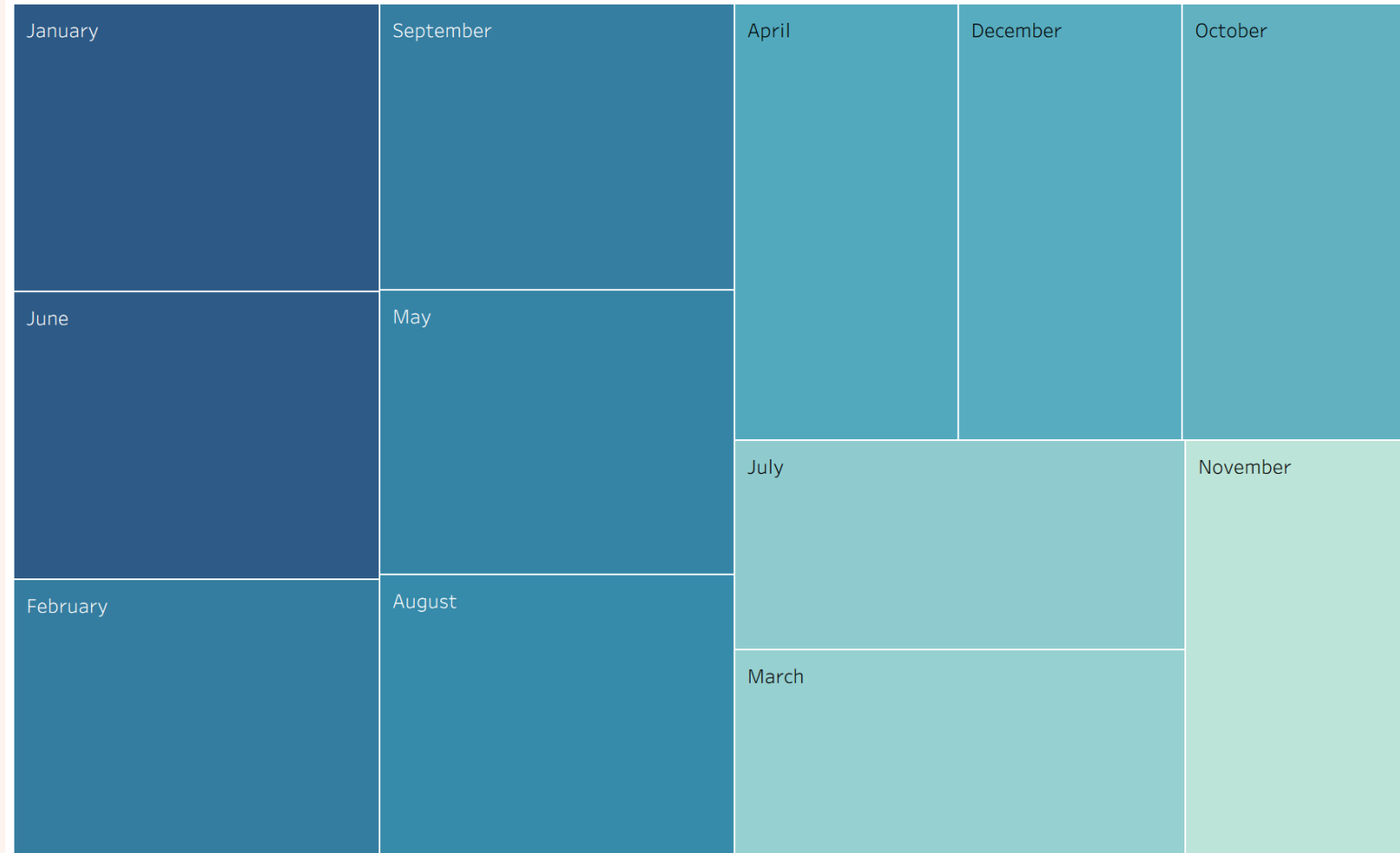
- Over recent years some numbers have been drawn more than others
- For example, number 29 was drawn the least in the Powerball in both 2022 and 2023
- We can use this information to pick numbers that have been picked often in the past in hopes that they will continue to be picked often

Powerball Lotto Numbers



- The average multiplier has been higher in January and June and lowest in November
- If we purchase the multiplier on a ticket for an extra \$1, we should do it in January or June since the multiplier may be higher then
- This could help us save money and only purchase the multiplier during a time when we think it is more likely to be larger

Average Multiplier Number by Month



Will you pitch in and buy multiple lottery tickets with me?

- By pooling our money together, we can increase our odds of winning (and splitting the prize) without each person having to spend a lot of money
- Pooling money together will allow us to increase odds of winning for a smaller cost
- Although lotteries are random, we can try and choose numbers that have appeared often in the past and use averages to pick our ticket numbers more strategically. This will not guarantee that we win however.